Joining data from multiple sources

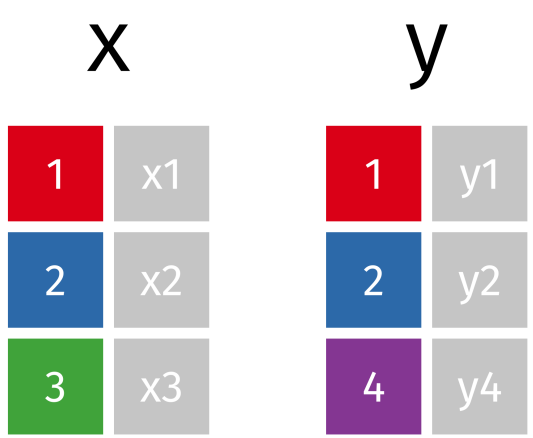
K Arnold, based on DSBox

A note on `mutate`

- Badly named. Think "add_computed_column".
- **DON'T** think of it operating on a variable ("mutate the ride's start time").
- **DO** think of it operating on a data frame ("add a column computed by flooring the start time)

Joining data frames

- I have a data frame x (e.g., Covid confirmed cases)
- I want extra information about things in x (e.g., population)
- Some other table, y, has that information. "Joins" let me look it up.
- Needs a key : what has to match. Must match exactly.



Graphics thanks to tidyexplain

Types of joins

What to do when things don't exactly line up? Start with all rows, but:

- **full** or **outer**: Leave blanks (`NA`) for mismatches
- **inner**: Drop rows with any mismatches
- **left** / **right**: Drop rows where one of the sides has a mismatch

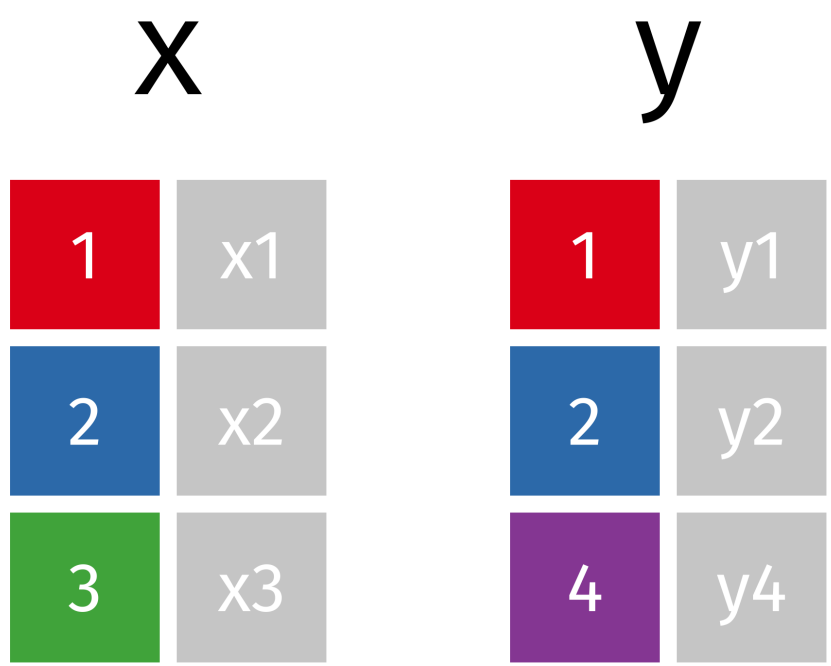
Setup

```
x
```

```
## # A tibble: 3 x 2
##     key xdata
##   <dbl> <chr>
## 1     1 x1
## 2     2 x2
## 3     3 x3
```

```
y
```

```
## # A tibble: 3 x 2
##     key ydata
##   <dbl> <chr>
## 1     1 y1
## 2     2 y2
## 3     4 y4
```

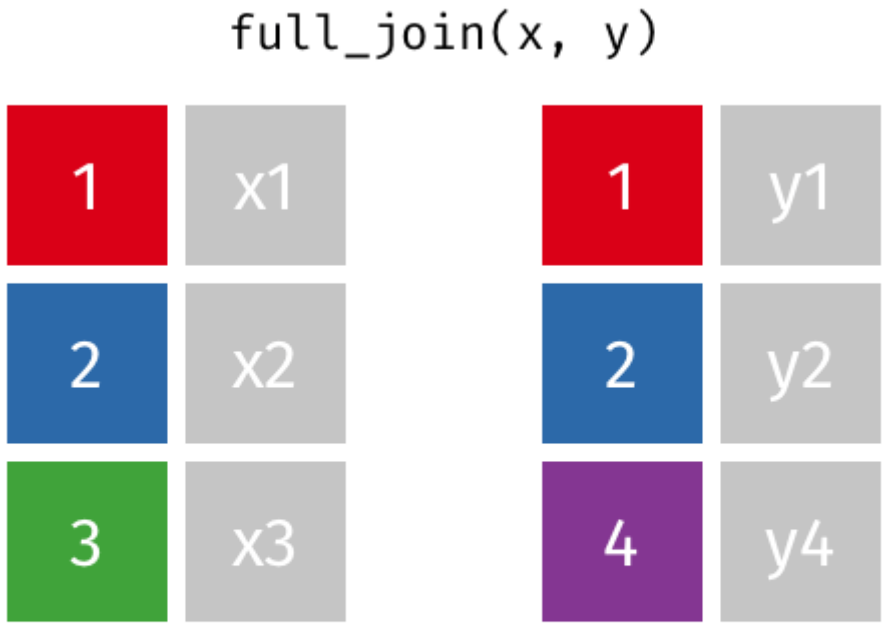


full_join()

All rows from both x and y. Leave NA for mismatches.

```
full_join(x, y, by = "key")
```

```
## # A tibble: 4 x 3
##     key xdata ydata
##   <dbl> <chr> <chr>
## 1     1 x1    y1
## 2     2 x2    y2
## 3     3 x3    <NA>
## 4     4 <NA>  y4
```

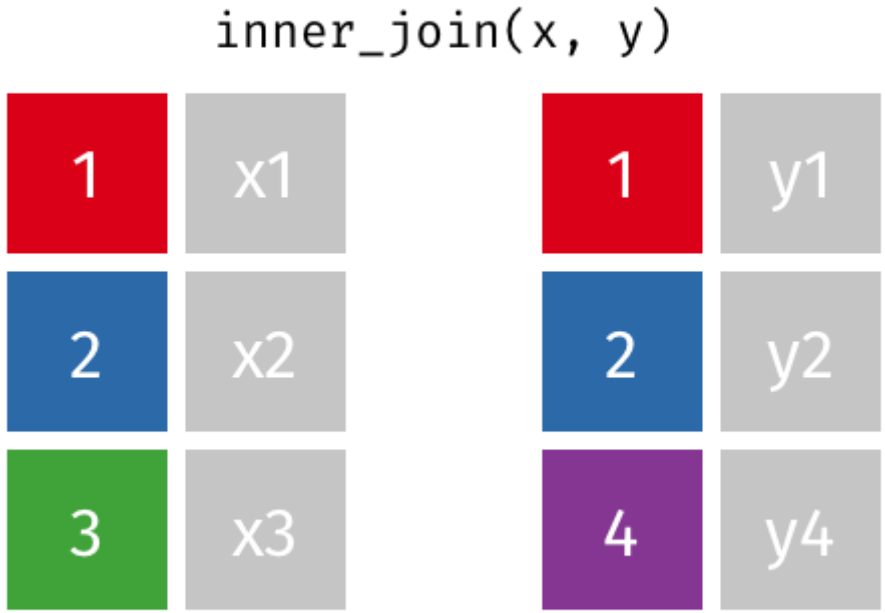


inner_join()

All matching rows. Drops mismatches.

```
inner_join(x, y, by = "key")
```

```
## # A tibble: 2 x 3
##     key xdata ydata
##   <dbl> <chr> <chr>
## 1     1 x1    y1
## 2     2 x2    y2
```

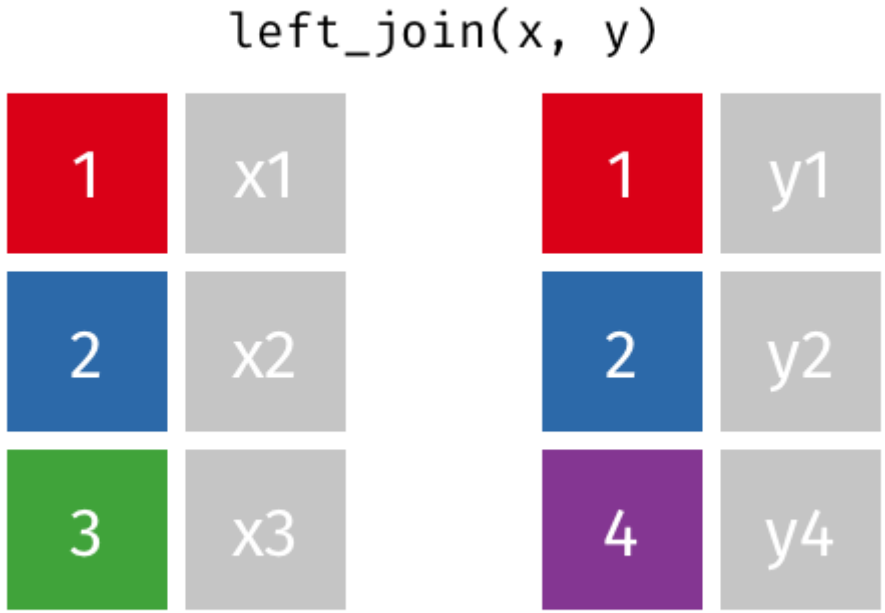


left_join()

All rows from x.

```
left_join(x, y, by = "key")
```

```
## # A tibble: 3 x 3
##     key xdata ydata
##   <dbl> <chr> <chr>
## 1     1 x1    y1
## 2     2 x2    y2
## 3     3 x3    <NA>
```



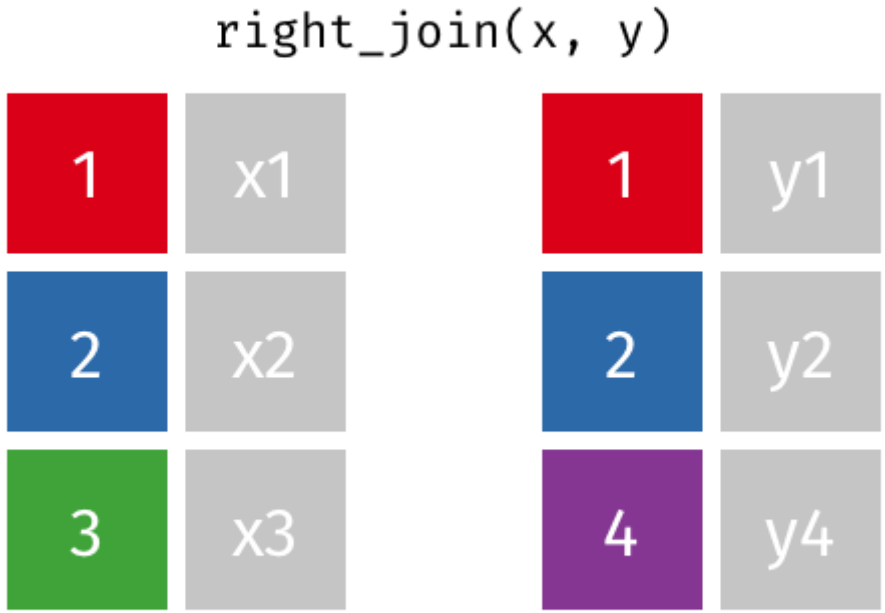
right_join()

All rows from y.

```
right_join(x, y)
```

```
## Joining, by = "key"
```

```
## # A tibble: 3 x 3
##     key xdata ydata
##   <dbl> <chr> <chr>
## 1     1 x1    y1
## 2     2 x2    y2
## 3     4 <NA>  y4
```



Summary

- `full_join()`: all rows from both `x` and `y`
- `inner_join()`: all *matching* rows from `x` where there are matching values in y. Multiple matches? Return all combinations.
- `left_join()`: all rows from x
- `right_join()`: all rows from y

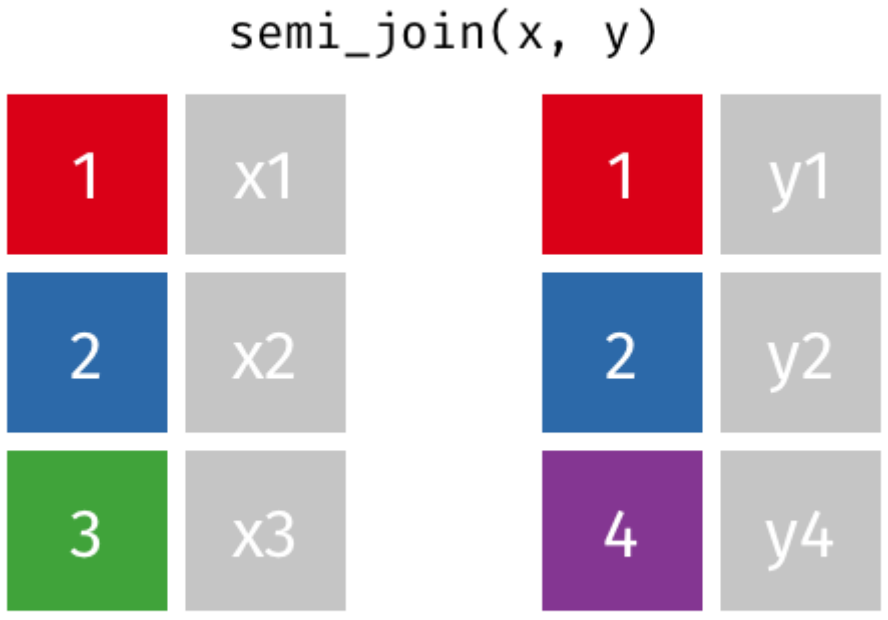
^ are called *mutating joins* (by analogy to the `mutate` verb). Sometimes useful: *filtering* joins and *nest* join:

- `semi_join()`: include a row from `x` only if there's some match in `y`
- `anti_join()`: include a row from `x` only if there's *no* match in `y`
- `nest_join()`: get bundles of all matching rows from `y` (most flexible)

semi_join()

```
semi_join(x, y, by = "key")
```

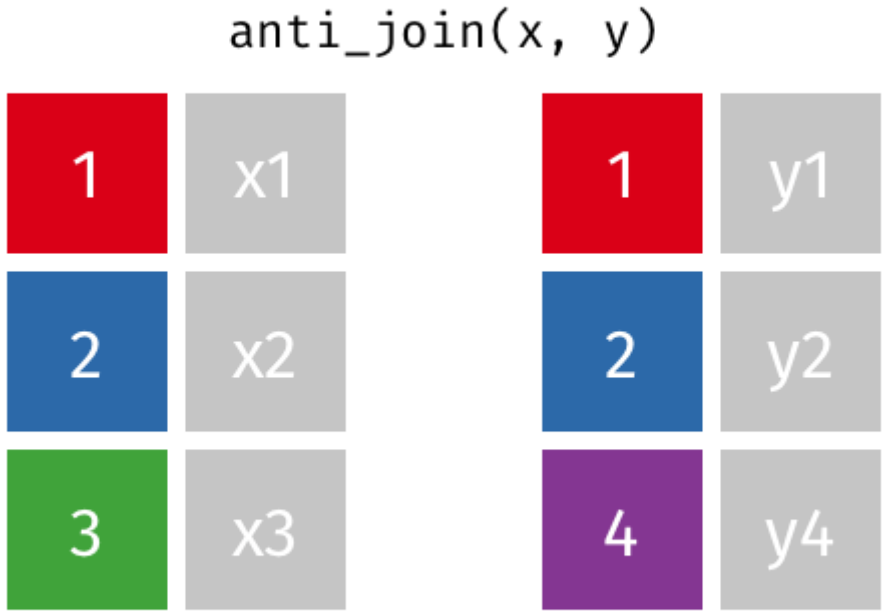
```
## # A tibble: 2 x 2
##     key xdata
##   <dbl> <chr>
## 1     1 x1
## 2     2 x2
```



anti_join()

```
anti_join(x, y, by = "key")
```

```
## # A tibble: 1 x 2
##     key xdata
##   <dbl> <chr>
## 1     3 x3
```



We want to keep all rows and columns from `confirmed_cases` and add a column for corresponding populations from `population`. Which join function should we use?